

Luciteria Collector Cabinet - Project Summary

Project Overview

Built a premium Shopify embedded app prototype that transforms element collecting into a subscription-driven experience with collection-aware assignment logic and duplicate prevention.

Status: Phase 2 Complete | Ready for Internal Review

Location: `/home/ubuntu/luciteria_collector_cabinet`

What Was Built

Phase 1: Core Foundation

- Collector Cabinet dashboard with summary cards
- Collection tracking by SKU (owned/missing/wishlisted/sent via subscription)
- Wishlist management
- Subscription management (skip, pause, cancel)
- Customer preferences (duplicate handling, budget filters)
- Admin operations dashboard
- Collection-aware assignment engine with duplicate prevention
- Mock database with 4 customer personas

Phase 2: Major Enhancements

- **Interactive Periodic Table UI** - All 118 elements, front-and-center, with status indicators
 - **Collection Type System** - Customer selects at signup (10mm, 25.4mm, 50mm, Lucite, or Ampoules); filtered periodic table
 - **Discount Monitoring** - Tracks subscription vs retail price; alerts admin if discount > 20%
 - **Advanced Billing** - Dual billing dates (signup day + 1st of month), handles month-end edge cases
 - **Grandfathering** - Price locks persist during subscription pause
 - **Enhanced Assignment Engine** - Out-of-stock sequence shifting, manual override capability, 4-month forecast
 - **Email Notifications** - Price change alerts (mock implementation)
 - **Webhook Structure** - Ready for Shopify order tracking
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Technology Stack

- **Framework:** Remix (server-side rendering)
- **Database:** Prisma ORM + SQLite (prototype); PostgreSQL for production
- **UI:** Shopify Polaris components
- **Authentication:** Mock session-based (production uses Shopify OAuth)
- **Styling:** Custom dark/light theme with premium aesthetic

Key Design Decisions

Decision	Approach	Rationale
Collections	Single collection per customer (forward-compatible)	Defer multi-collection; design for future expansion
Profitability	Track discount % from retail; alert if > 20%	Simple, actionable; avoids complex margin optimization
Billing	Signup day + 1st of month with edge case handling	Customer-friendly, clear rhythm; handles month-end edge cases
Out of Stock	Shift sequence to next eligible item	Prevents subscription lapses; maintains rhythm
Admin Approval	Auto-approve assignments; optional manual override	Reduces friction; gives admin control when needed
Periodic Table	Show all 118 elements; grey out unavailable	Collection completion metric is always /118, not /20
Grandfathering	Clock continues during pause	Protects locked pricing for subscribers
Subscription	Standalone (not Recharge/Seal)	Full control over duplicate-prevention logic; no third-party fees

Current Data

- **35 products** from actual Luciteria catalog
- **6 customer personas** covering all 5 collection types and edge cases
- **118 elements** in periodic table (complete periodic table structure)
- **Realistic scenarios:** Completionists, newcomers, high-discount situations, pause/resume, price grandfathering

How to Run

```
cd /home/ubuntu/luciteria_collector_cabinet
npm install
npm run dev
```

Visit: `http://localhost:3000`

Test URLs:

- Dashboard: `/app/cabinet`
- Periodic Table: `/app/cabinet` (visible on dashboard)
- Onboarding: `/app/cabinet/onboarding`
- Admin Ops: `/app/admin/operations`

Customer personas: Add `?customer=cust_001` (or 002-006)

Key Files

- **Assignment Logic:** `/app/lib/assignment-engine.server.js`
- **Billing Logic:** `/app/lib/billing.server.js`
- **Periodic Table:** `/app/components/PeriodicTable.jsx`
- **Schema:** `/prisma/schema.prisma`
- **Documentation:** `ARCHITECTURE.md` , `ASSIGNMENT_LOGIC.md` , `PRODUCTION_ROADMAP.md`

Fixed Issues (Recent)

1.  Admin operations now auto-approve (no manual approval required)
2.  Collection progress always shows “X / 118” (not X / available products)
3.  Periodic table rendering fixed and displays on customer dashboard

What's Next for Production

7-12 Week Production Roadmap:

- Real Shopify API integration (Customers, Products, Subscriptions)
- PostgreSQL database migration
- Shopify OAuth authentication
- Webhook setup for order automation
- Email service integration
- Refund/dunning management
- Shopify app publishing

See `PRODUCTION_ROADMAP.md` for detailed breakdown.

Key Assumptions

- **Precious metals excluded** from normal assignments (Au, Pt, Pd, Rh, Ir, Os, Ru, Re)
- **No COAs offered** (per business policy)
- **Out-of-stock items** shifted in sequence, not skipped
- **Collection types locked** at signup in Phase 2 (future phase allows changes)

- **Billing always resolves** to day-of-signup and 1st-of-month
 - **Admin can override** assignment sequence but cannot extend grandfathering window
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For Next Conversation

Reference this summary when:

- Requesting tweaks to prototype
- Planning production roadmap
- Scaling to multiple collections
- Integrating with real Shopify
- Adding new features or workflows

Project remains stable and fully functional. All Phase 2 enhancements tested and working.